Waste Heat Recovery Pathways for AI Data Centers: Emerging Opportunities, Decarbonization Potential, and Future Perspectives

Yingbo Zhang

Research Scientist, University of California, Santa Barbara

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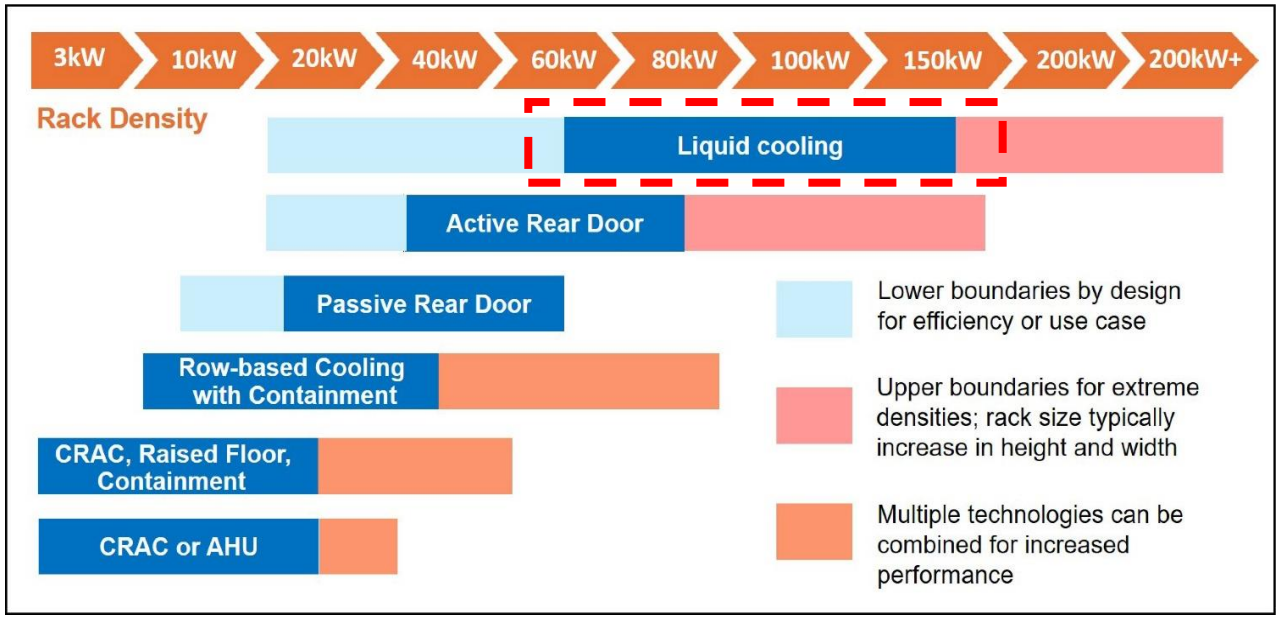
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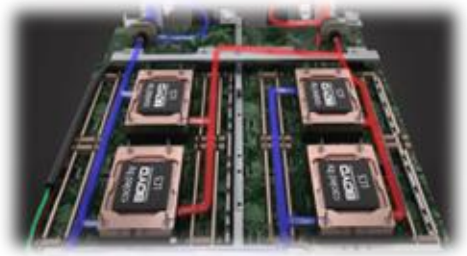
1. Background

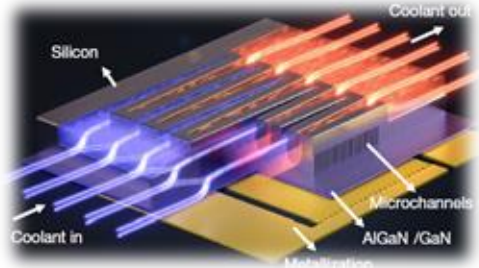
❑ AI-focused data center











- As rack power density increases, from 7 kW in 2021 to **130 kW** in 2026, **19-fold increase**
- Air-cooled systems are insufficient, a shift toward liquid cooling solutions
- Circulate coolant through a metal plate in direct contact with the heat source (CPU/GPU)
- Allow outlet liquid temperatures up to **50-60℃**, high-grade waste heat

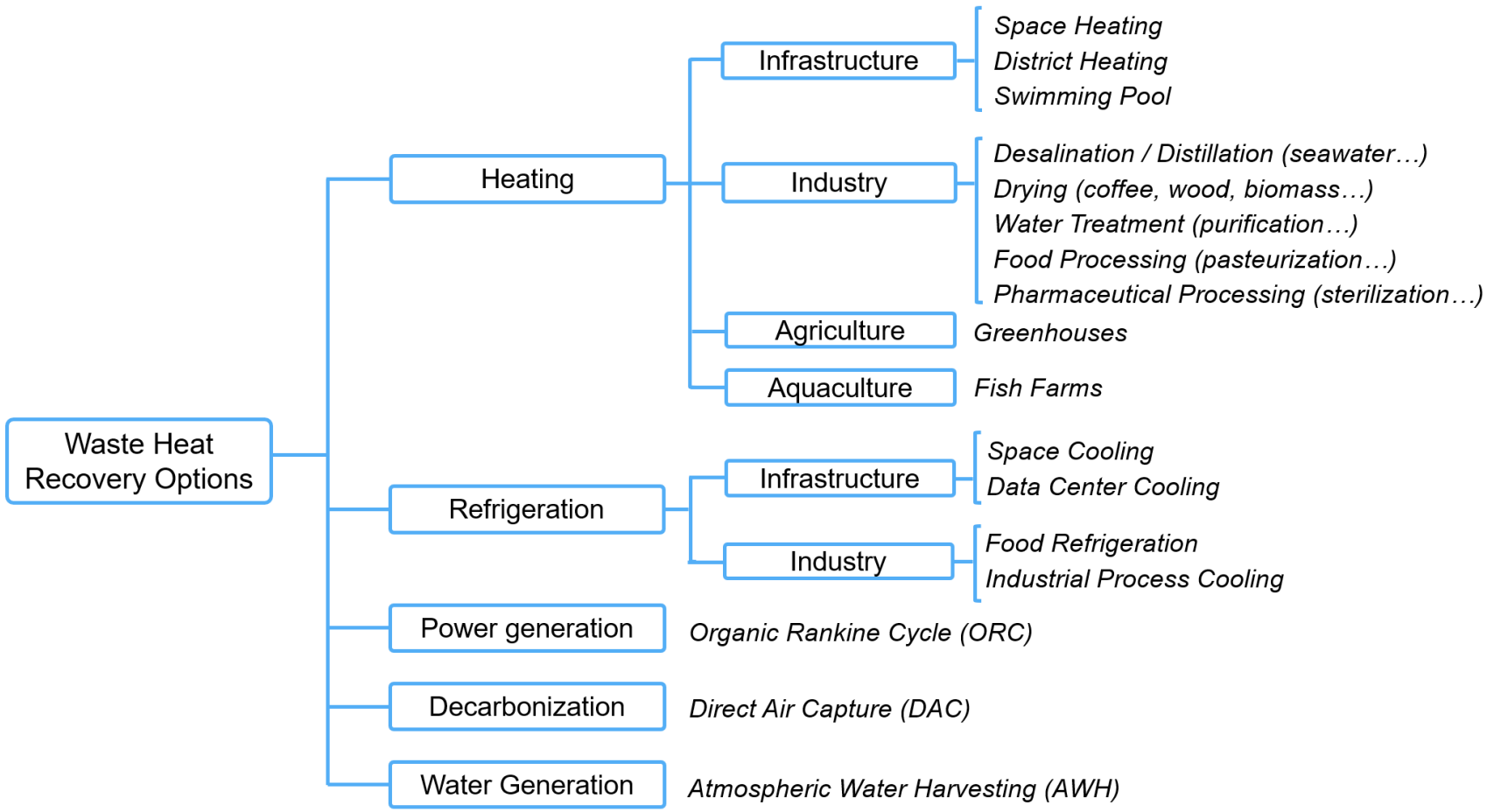
2. Waste Heat Recovery Options

Five main categories

- Heating
- Cooling
- Power generation
- Decarbonization
- Water Generation

Main Application Areas

- Infrastructure
- Industry
- Agriculture
- Aquaculture
- Environmental Applications



3. Case Studies and Literature Review

❑ Case summary of data centers with waste heat recovery

- Further technical details are provided in the Supplementary Information (SI).

Data center	Operation	Location	IT load	Tin	Tout	Cooling methods
Lichtenberg II high-performar	TU Darmstadt	Darmstadt, Germany	620 kW	45-50 °C	55-60 °C	Cold plate
Data center at Wallotstraße, C	Cloud&Heat	Dresden, Germany			50-60 °C	Cold plate
Cloud&Heat Edge	Cloud&Heat	Frankfurt, Germany	250 kW		60 °C	Cold plate
Greifswald New Data Centre	State of Mecklenburg -	Greifswald, Germany	340 kW waste heat		60 °C	Cold plate
Horeka, KIT, High performance	KIT, Lenovo	Karlsruhe, Germany	750 kW	40 °C	45°C (max 55	Cold plate
LRZ (Leibniz Supercomputing	Bavarian Academy of S	München - Garching	4 MW	50 °C	60 °C	Cold plate
Hunter, HLRS, University of Stu	the University of Stuttg	Stuttgart, Germany	3.5 MW	25 °C	35 °C	Cold plate
Levante, DKRZ (Deutsches Klin	University of Hamburg	Hamburg, Germany	2.5 MW	40 °C	48 °C	Liquid Cooling
JUPITER, JSC (Jülich Supercom	Jülich Supercomputing	Jülich, Germany	4.5 MW	36 °C	> 40 °C	Cold plate
TUD (Technische Universität D	Technische Universität	Dresden, Germany	>500 kW	35 °C	48 °C	Cold plate
iDataCool (Univ. Regensburg	University of Regensbu	Regensburg, Germar	60 kW	55-65°C	60–70°C	Cold plate
University of Oldenburg HPC	University of Oldenbur	Oldenburg, Germany			45°C	Cold plate
Data center campus at the ICE	OVHcloud	Limburg, Germany	35 MW	30 °C		Cold plate
BlockHeating	EcoDataCenter, WA3RI	Venlo, Netherlands	200 kW		65 °C	Cold plate
Switch Datacenters	Switch	Woerden, Netherlanc	800 kW		40-60 °C	Liquid Cooling
Data Center Bytesnet	Bytesnet	Groningen, Netherlar	200 kW		50°C-55°C	Immersion cooling
3D-Animation Studio	TeamTO	Paris, France			27 °C	Immersion cooling
Paris PA10 (Equinix)	Equinix	Paris, France			28 °C	Liquid cooling
Deep Green Data Center	Deep Green	Exmouth, Great Britai	28 kW		30 °C	Immersion cooling

3. Case Studies and Literature Review

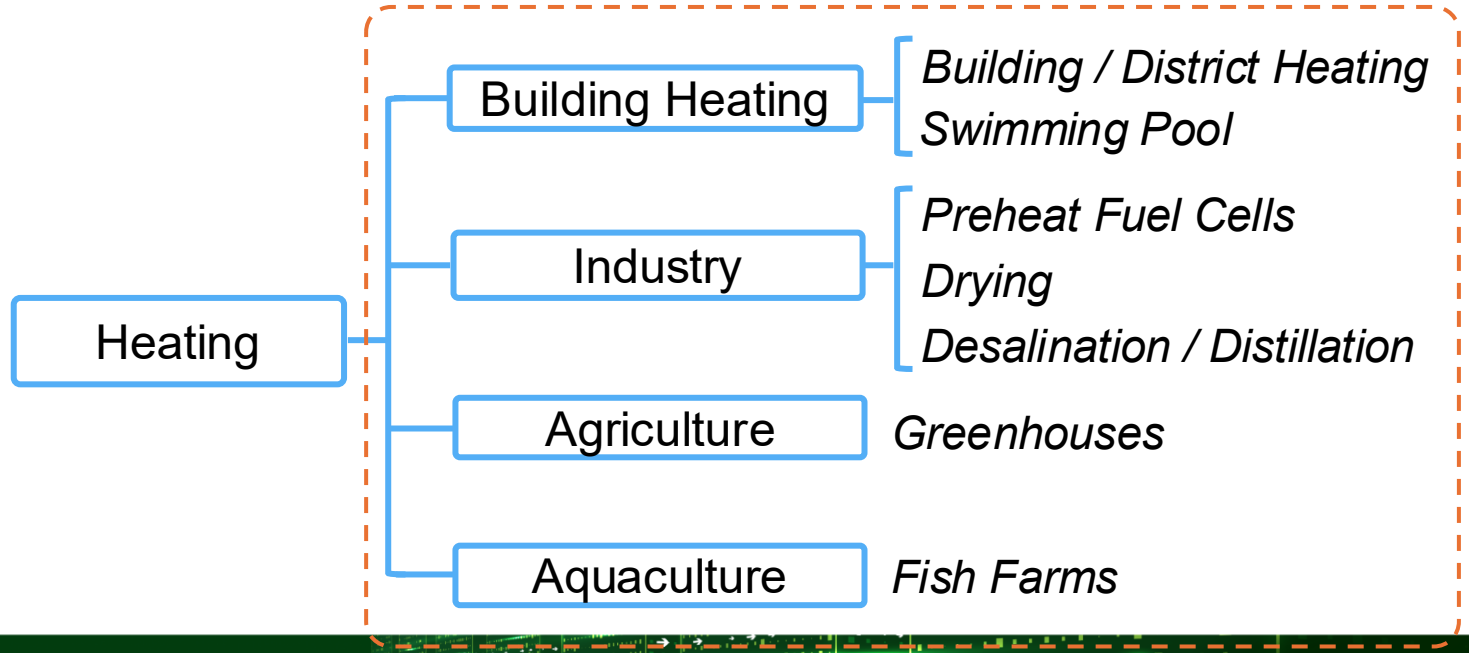
❑ Scientific research and company announcement

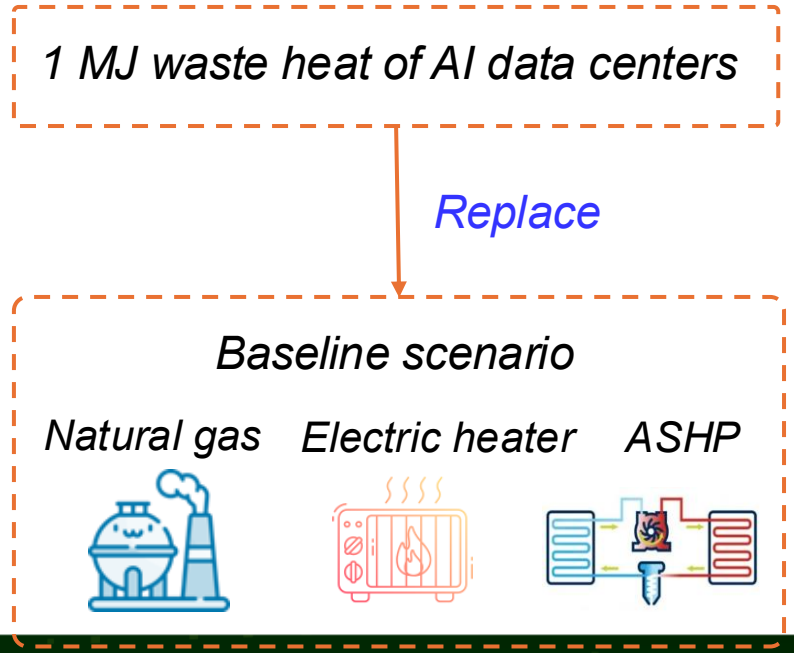
Paper	Data center	Year	Location	Cooling	Option
Techno-economic analysis c	1	2025	USA	Liquid cooled	Power generation
Study of a Lab-Scale Organi	2	2020	USA	Liquid cooled	Power generation
Turning Data Center Waste H	3	2024	Italy	Liquid cooled, Air coole	Power generation
Comparative study for waste	4	2024	China	Liquid cooled	Power generation
A techno-economic evaluati	5	2022	Canada	Liquid cooled	Power generation
Experimental and numerical	6	2022	Italy	Liquid cooled	Power generation
The viability of ultra low tem	7	2017	USA	Liquid cooled	Power generation
Organic rankine cycle as a w	8	2018	USA	Liquid cooled	Power generation
Multi-mode operation and t	9	2022	China	Liquid cooled	Power generation
Thermo-Economic Potential	10	2025	China	Liquid cooled	Power generation
Performance Analysis of a C	11	2024	China	Liquid cooled	Power generation
An Integrated Data Center E	12	2025	China	Liquid cooled	Power generation
Research and Application Pr	13	2026	China	Liquid cooled	Refrigeration
Switchable multistage absor	14	2025	China	Liquid cooled	Refrigeration
High-power-density adsorp	15	2024	China	Liquid cooled	Refrigeration
Development and experimer	16	2024	China	Liquid cooled	Refrigeration
Application analysis of adso	17	2021	China	Liquid cooled	Refrigeration
Waste heat recovery in a dat	18	2021	Canada	Liquid cooled	Refrigeration
Thermodynamic feasibility o	19	2012	USA	Liquid cooled	Refrigeration

4.1 Waste Heat Recovery for Heating Applications

❑ A Generalized Comparative Analysis

- **Direct Heat Reuse**: Assumes that 1 MJ of recovered waste heat directly replaces baseline heating energy (natural gas or electricity).
- **Limitations**: Supply–demand matching is not considered in this simplified calculation and will be addressed in future technical paper.
- **Carbon Benefits**: Eliminates emissions by displacing traditional fuel combustion and grid power.
- **Liquid cooling Advantage**: High-grade waste heat allows for direct application without heat pumps.



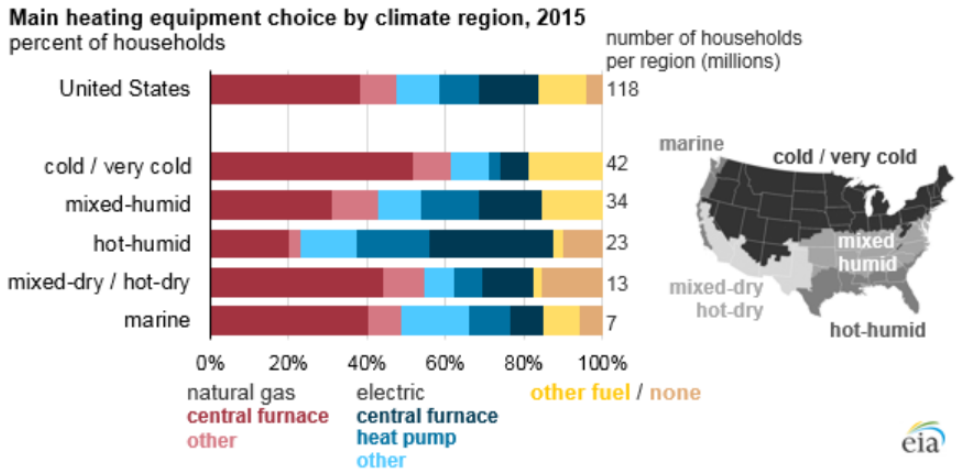


4.1 Waste Heat Recovery for Heating Applications

❑ Building heating systems in the US

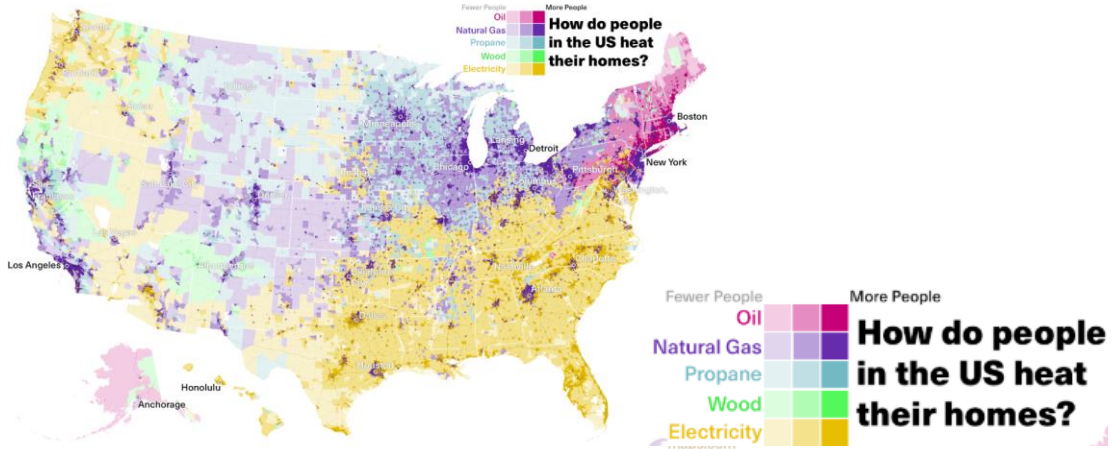
Building heating systems

- Natural Gas Furnace (common)
- Fuel-oil boiler (Northeast, Boston)
- Propane (rural areas)
- Wood heating (rural areas)
- Air Source Heat Pump (grow rapidly)
- Electric Heating (auxiliary)



Source: U.S. Energy Information Administration, 2015 Residential Energy Consumption Survey
Note: Climate regions based on Department of Energy Building America climate regions.

[1]



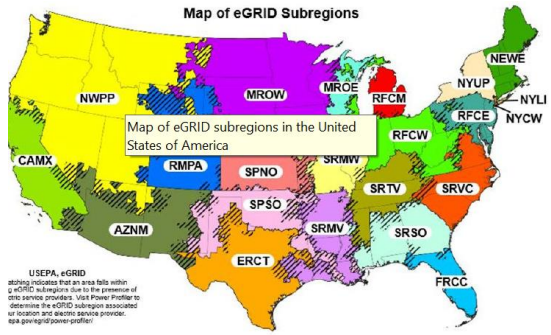
[1] https://www.eia.gov/todayinenergy/detail.php?id=30672

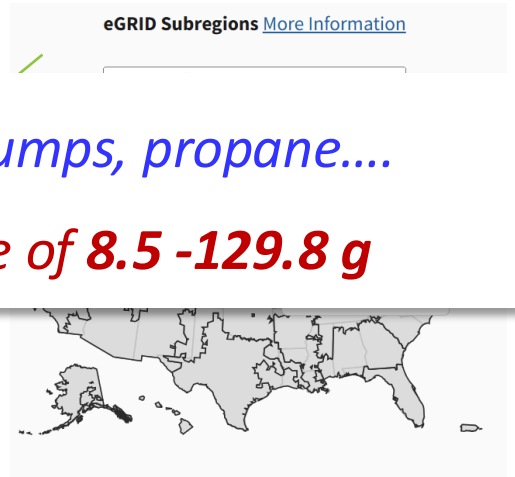
[2] https://www.maps.com/home-heating-fuels/

4.1 Waste Heat Recovery for Heating Applications

❑ Carbon emissions coefficients of building heating systems in the US

Heating system	Typical COP/ efficiency	Fuel / grid CO_2 intensity
Natural Gas	90%	0.201 $kgCO_2/kWh$ [1]
Fuel-oil boiler	85%	0.298 $kgCO_2/kWh$ [1]
Propane	90%	0.239 $kgCO_2/kWh$
Wood	80%	0.413 $kgCO_2/kWh$
Electric heater	COP 1.0	0.11-0.67 $kgCO_2/kWh$ (Grid) [5]





- *Baseline scenarios: natural gas, electric heaters, air source heat pumps, propane….*
- *Reuse 1 MJ waste heat for direct building heating can reduce CO_2e of **8.5 -129.8 g***

Natural gas 0.181 $kgCO_2/kWh$; Fuel Oil 0.253 $kgCO_2/kWh$[1]

Coal 0.328 $kgCO_2/kWh$; Propane 0.215 kgCO2/kWh; Wood: 0.330 $kgCO_2/kWh$ [4]

[1] https://www.eia.gov/environment/emissions/co2_vol_mass.php
[2] https://www.iso-ne.com/static-assets/documents/100016/2023-air-emission-report-20241016.pdf?utm
[3] https://www.pnnl.gov/main/publications/external/technical_reports/PNNL-37127.pdf?utm
[4] https://www.pfpi.net/carbon-emissions/
[5] https://www.epa.gov/egrid/summary-data?utm

Official carbon intensity databases

4.1 Waste Heat Recovery for Heating Applications

❑ Waste heat for greenhouse heating: representative cases

- **Temperature match:** Data center waste heat, temperature match, greenhouse applications, fruit and vegetable.
- **Case 1:** Equinix company utilizes data center waste heat for sustainable rooftop farming in France [1].
- **Case 2**: QScale company utilizes data center waste heat to grow more than 80,000 tons tomatoes per year in Canada [2].
- **Case 3**: A data center use data center waste heat to grow tropical fruit in Sweden [3].
- **Case 4**: A Terra Ventures data center uses waste heat to grow strawberries and tomatoes in California [4].

➢ *Baseline scenarios: natural gas, electric heaters, air source heat pumps*

➢ *Reuse 1 MJ waste heat for greenhouse heating can reduce CO_2e of **8.5 -185.0 g***







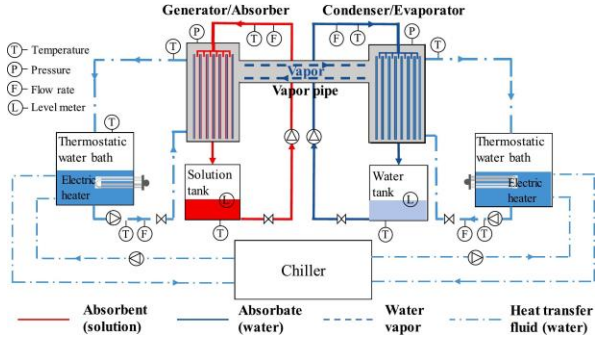
[1] https://www.grozine.com/2023/03/15/data-centre-powered-greenhouse/
[2] https://corporateknights.com/food/this-company-growing-food-using-heat-from-data-centres/
[3] https://www.youtube.com/watch?v=pHg1MGe4tMM
[4] https://www.linkedin.com/posts/david-rockhill-3860542_strawberry-fields-for-data-as-any-data-activity-7407483443300945920-qZ4e

4.2 Waste heat recovery for refrigeration

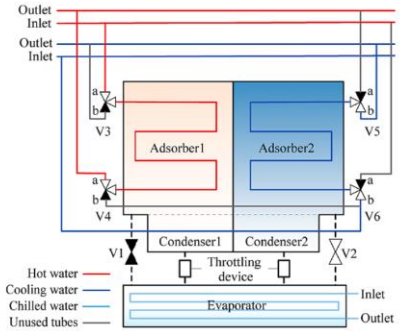
❑ Principles of absorption/adsorption refrigeration

- **Thermal Driving Force**: Unlike traditional systems that use compressors, both systems use heat (waste heat, solar, or steam) as the primary energy source.
- **The Sorption Cycle**: A "sorbent" (liquid or solid) captures refrigerant vapor at low pressure/temperature and releases it at high pressure/temperature when heated, creating a continuous cooling effect.

Feature	Absorption (LiBr-Water)	Adsorption (Silica Gel-Water)
State of Sorbent	Liquid (e.g., Lithium Bromide solution)	Solid (e.g., Silica Gel or Zeolite)
Mechanism	Chemical/Physical dissolving into the bulk liquid	Physical bonding to the surface of the solid
Driving Temp	Higher (>80°C typically required).	Lower (starts as low as 50°C).



Schematic of absorption refrigeration cycle



Schematic of adsorption refrigeration cycle

4.2 Waste heat recovery for refrigeration

❑ Question: Why not use heat pumps to upgrade waste heat for refrigeration?

- **Low overall system COP**: the combined COP of the heat pump and absorption chiller is lower the baseline of a standard electric chiller, higher energy consumption
- **Higher investment**: higher capital expenditure without delivering operational energy savings

Metric	Heat pump + absorption	Traditional chiller
Same cooling Output	500 kW	500 kW
Electricity Input	**178.6 kW**	**100 kW**
Net COP	2.8	5.0
Waste Heat Use	535.7 kW	0 kW

Parameters

- **Target cooling load**: 500 kW
- Heat pump COP (60℃ to 90℃): 4.0
- Traditional Chiller COP: 5.0
- Absorption chiller COP: 0.7 (at 90℃)

The comparison assumes the same 500 kW refrigeration load. The 535.7 kW is data center waste heat that would otherwise be rejected to the environment.

➢ Directly use waste heat (50-60℃) for adsorption/absorption refrigeration cycles

[1] Cui Z, Du S, Wang R. Switchable multistage absorption chiller for ultra-low-grade waste heat driven cooling in data centers[J]. Applied Energy, 2025, 398: 126407.
[2] Zenggang S, Baosheng L, Yunren S, et al. Research and Application Progress of Sorption-Based Energy and Water-Saving Technologies in Data Centers[J]. Journal of Refrigeration, 2025 (null): 1-14.

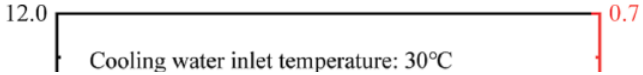
4.2 Waste heat recovery for refrigeration

❑ Adsorption cooling performance

- **On-site cooling**: directly use waste heat (50-60℃) to drive adsorption chiller (driving temperature: as low as 50℃, e.g. Silica Gel-Water)
- **Chiller performance**: Experimental results, adsorption chiller COP is around 0.45-0.5 [1,2]
- **Sample**: InvenSor LTC 90 e plus adsorption chiller [3], COP is around 0.5-0.6, **0.5** is assumed



Schematic of an absorption refrigeration system



Experimental data of an adsorption chiller



Performance data of adsorption chillers

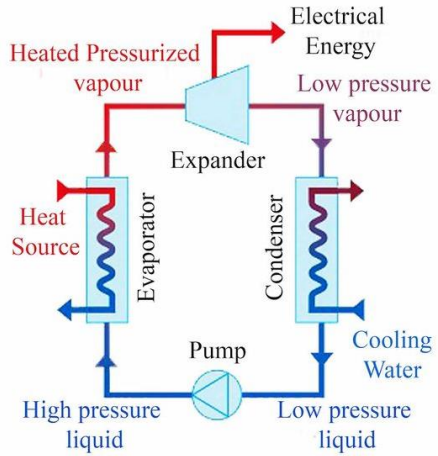
- *Baseline scenarios: conventional compressor-based chillers, typical COP is around 5*
- *Reuse 1 MJ waste heat for refrigeration can reduce CO_2e of **1.8 -14.9 g***
- A 30 MW data center, waste heat recovery for onsite air-side cooling can reduce approximately 1,362–11,277 tons of CO_2 emissions annually—equivalent to planting over half a million trees or eliminating more than 11,000 transatlantic flights.

[1] Gupta R, Puri I K. Waste heat recovery in a data center with an adsorption chiller: Technical and economic analysis[J]. Energy Conversion and Management, 2021, 245: 114576.
[2] Du S, Cui Z, Wang R Z, et al. Development and experimental study of a compact silica gel-water adsorption chiller for waste heat driven cooling in data centers[J]. Energy Conversion and Management, 2024, 300: 117985.
[3] https://www.aaasaveenergy.com/products/001/pdf/InvenSor_LTC90_e_plus_datasheet.pdf

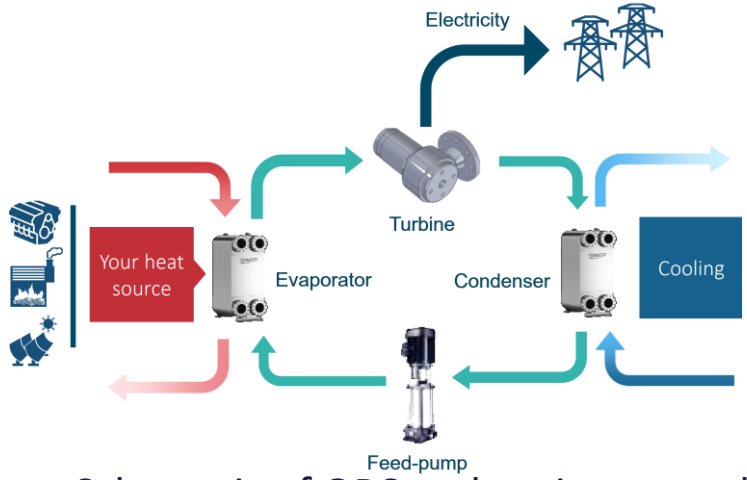
4.3 Waste heat recovery for power generation

❑ Principles of Organic Rankine Cycle (ORC) power generation

- **Waste heat utilization:** ORC converts medium-temperature waste heat (typically 80–300°C) into electricity.
- **Organic working fluid**: ORC uses organic fluids with low boiling points (e.g., R245fa, pentane)
- **Thermodynamic cycle**: The working fluid is evaporated by waste heat, expanded through a turbine to generate power, condensed, and then pumped back to repeat the cycle.
- **Electricity generation**: The turbine drives a generator to produce electricity from recovered thermal energy.



Schematic of ORC cycle



Schematic of ORC cycle using waste heat

4.3 Waste heat recovery for power generation

❑ Why Not Use Heat Pumps to Upgrade Waste Heat?

- **Negative Net Power Output**: Electricity consumed by the heat pump offsets ORC power generation.
- **Literature validation:** Ref [1] shows that using waste heat directly for ORC saves more energy than using a heat pump.

1. Electricity input of heat pumps (W_{HP})

$$W_{HP} = \frac{500\ kW}{COP-1} = 166.7\ \text{kW}$$

2. ORC power output (W_{ORC})

$$W_{ORC} = (500 + 166.7)\ \text{kW} \times 15\% = 100\ \text{kW}$$

3. Net power

$$\text{Net power} = 100\ \text{kW} - 166.7\ \text{kW} = -66.7\ \text{kW}$$

Parameters

- Waste heat: 500 kW
- Heat pump COP (60℃ to 90℃): 4.0
- ORC efficiency: 15% (at 90℃)

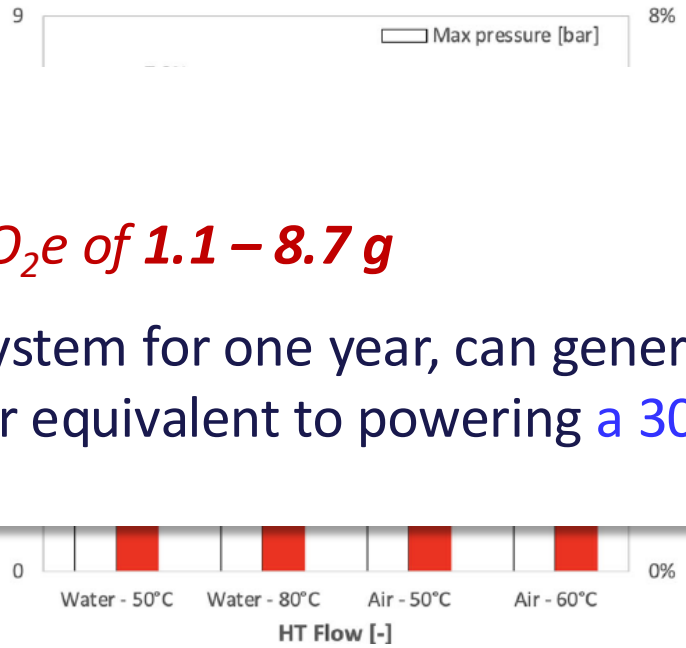
➢ Directly use data center waste heat (50-60℃) for ORC power generation (low efficiency) [2]

[1] Zhou X, Xin Z, Tang W, et al. Comparative study for waste heat recovery in immersion cooling data centers with district heating and organic Rankine cycle (ORC)[J]. Applied Thermal Engineering, 2024, 242: 122479.
[2] Araya S, Wemhoff A P, Jones G F, et al. Study of a lab-scale organic rankine cycle for the ultra-low-temperature waste heat recovery associated with data centers[J]. Journal of Electronic Packaging, 2021, 143(2): 021001.

4.3 Waste heat recovery for power generation

❑ Power generation efficiency

- **Electrical efficiency**: 3.7%-4.7%, when driven by waste heat at 50-60℃ [1, 2].
- **Auxiliary equipment**: additional water pumps, typically representing 10%–15% of total power generation [1].
- **Net Power**: Power generation – additional power input



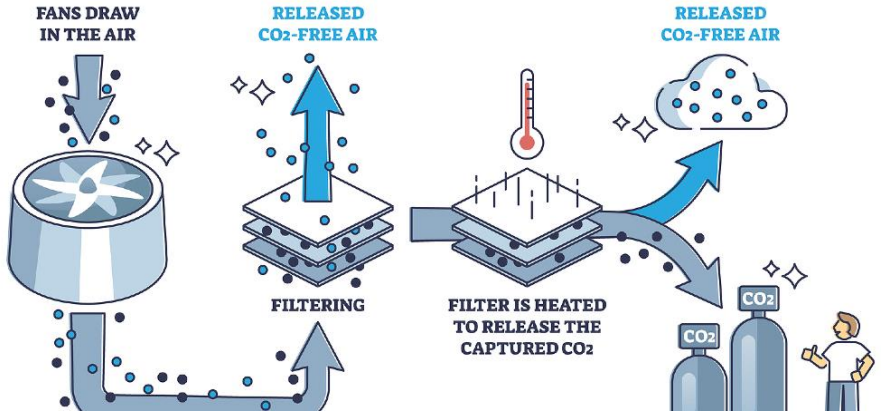
- *Baseline scenarios: using electricity from the power grid*
- *Reuse 1 MJ waste heat for power generation can reduce CO_2e of **1.1 – 8.7 g***
- A 30 MW data centers, waste heat for power generation system for one year, can generate 8.4 GWh, can supply about 780 households for one year, or equivalent to powering a 30 MW data center for about half a day.

[1] Araya S, Wemhoff A P, Jones G F, et al. Study of a lab-scale organic rankine cycle for the ultra-low-temperature waste heat recovery associated with data centers[J]. Journal of Electronic Packaging, 2021, 143(2): 021001.
[2] Corigliano O, Algieri A, Fragiacomo P. Turning data center waste heat into energy: a guide to organic Rankine cycle system design and performance evaluation[J]. Applied Sciences, 2024, 14(14): 6046.

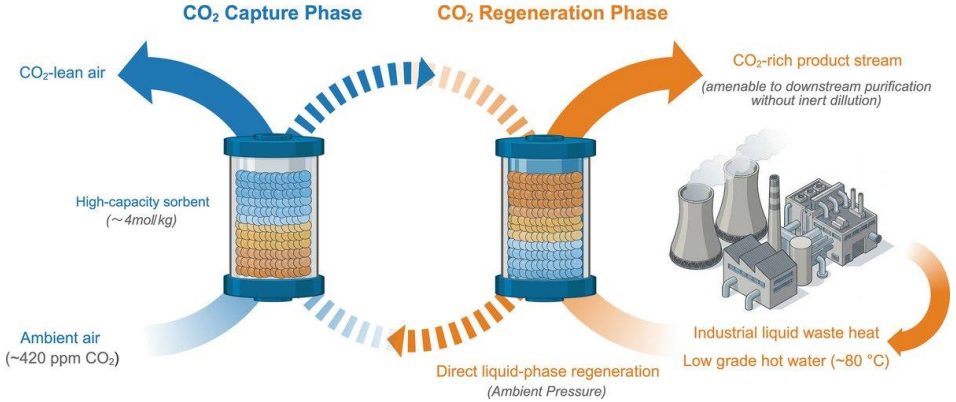
4.4 Waste heat recovery for direct air capture

❑ Principles of waste heat-driven direct air capture (DAC)

- **Sorbent Adsorption**: Use large fans to pull ambient air through a contactor, where CO_2 is chemically captured by a solid sorbent that acts like a chemical magnet.
- **Thermal demand:** DAC processes require significant amounts of heat (80°C-120°C) to release the captured CO_2 from the filters (sorbent) (thermal energy, break the chemical bonds between captured CO_2 and the filter).
- **Data center waste heat**: Heat pumps are required to raise waste heat temperature from 40-50°C to 80-120°C range necessary for DAC technology.



Schematic of direct air capture



Schematic of direct air capture using waste heat

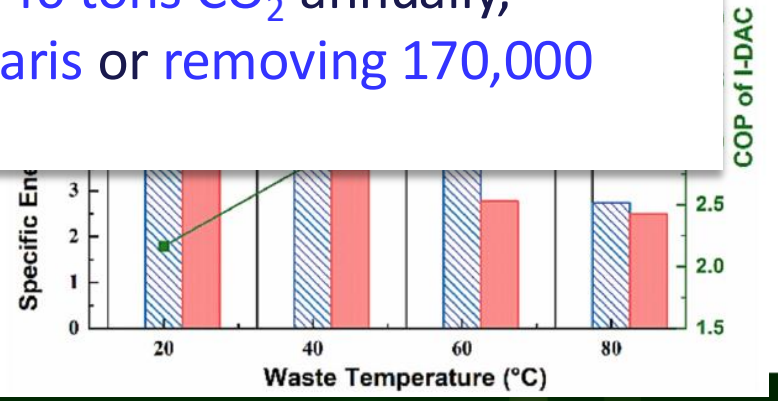
4.4 Waste heat recovery for direct air capture

❑ Referenced results from literature

- **Temperature Upgrade**: In Ref [1], heat pumps are used to raise waste heat (50-60°C) temperature to meet DAC requirements (100°C).
- **Onsite Energy "Penalty"**: Consider the additional power consumption of heat pumps, circulating pumps
- **Data from literature**: Auxiliary equipment electricity demand and the total thermal energy required for carbon capture.

- ➢ *Reuse 1 MJ waste heat for direct air capture can reduce CO_2e of **116.0 – 226.5 g***
- ➢ A 30 MW data center has the potential to reduce 87,796 – 171,440 tons CO_2 annually, equivalent to the carbon impact of planting a forest the size of Paris or removing 170,000 transatlantic flights.

APDES-NFC	humid	TVSA	120	8.6–10.1	1.1–1.3	~200	[25]
Lewatit	humid			10.1–11.8	0.6–1.3		
	humid	TVSA		0	9.08	151.3 (OPC)	
PEI67/Mg0.55Al–O	humid	H-DAC	100		3.53	58.8 (OPC)	This work
	dry	I-DAC		4.03	1.92	32.0 (OPC)	
	humid				2.77	46.2 (OPC)	

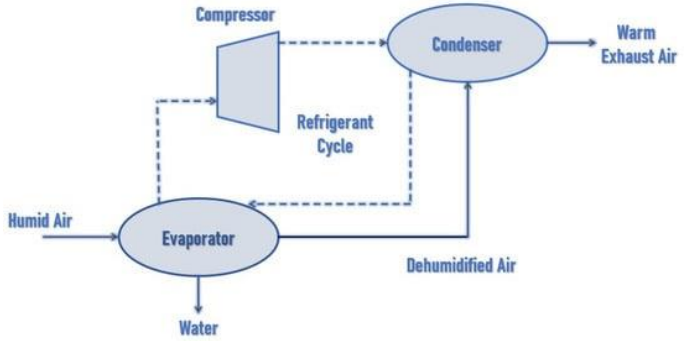


[1] Ge B, Zhang M, Hua B, et al. Innovative process integrating high temperature heat pump and direct air capture[J]. Applied Energy, 2024, 355: 122229.
[2] Fasihi M, Efimova O, Breyer C. Techno-economic assessment of CO2 direct air capture plants[J]. Journal of cleaner production, 2019, 224: 957-980.

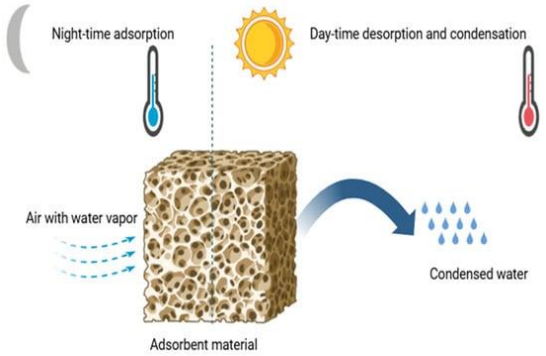
4.5 Waste heat recovery for extracting water from air

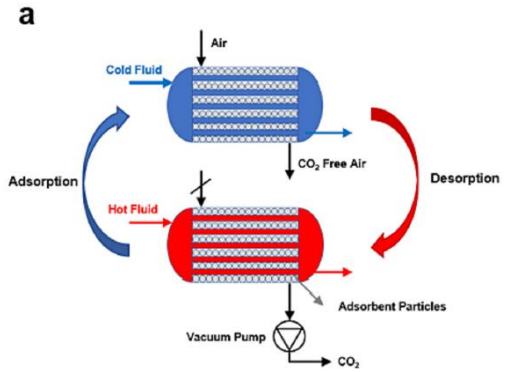
❑ Atmospheric water harvesting (AWH): extracting water from air

- **Main Technologies**: Condensation-based systems, Sorption-based systems (e.g., silica gel, driving temp. 80-120 °C [1]).
- **Operating Principle**: Water is condensed by cooling air below the dew point; captured by hygroscopic materials and later released through heating
- **Applications**: Used for drinking water supply, remote communities, military operations, and arid regions with limited freshwater resources.



Condensation-based systems





Sorption-based systems

[1] https://www.iea.org/energy-system/carbon-capture-utilisation-and-storage/direct-air-capture

4.5 Waste heat recovery for extracting water from air

❑ Comparison between condensation-based and sorption-based methods

Feature	Condensation-based	Sorption-based
Principle	Cooling air below dew point	Sorbent captures and releases vapor
Energy source	Electricity	Waste heat
Technology maturity	Mature, traditional	Emerging, trending
Energy Requirement	0.3-2 kWh/L (strongly dependent on humidity and temperature)	0.3-1.7 kWh/L
Limitation	Strongly dependent on high humidity	Complex system and material constraints
Advantage	Mature technology; simple system;	**Can operate in low humidity**; can utilize low-grade heat (e.g., solar or waste heat)

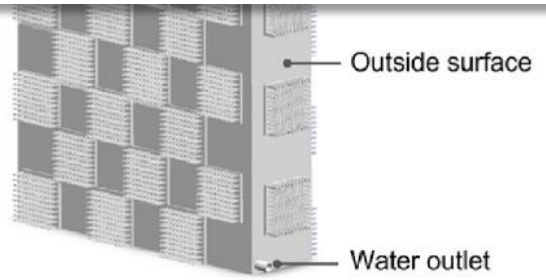
- **Data center waste heat**: data center waste heat can be used for extracting water from air by sorption-based technologies, enabling water production in arid regions where condensation-based systems are less effective.
- **Scope**: mainly compare their energy consumption on **low humidity conditions**
- **Sorption based AWH**: data center waste heat 50–60 °C, insufficient for effective desorption, a Heat Pump is needed to raise the temperature to 100-120°C

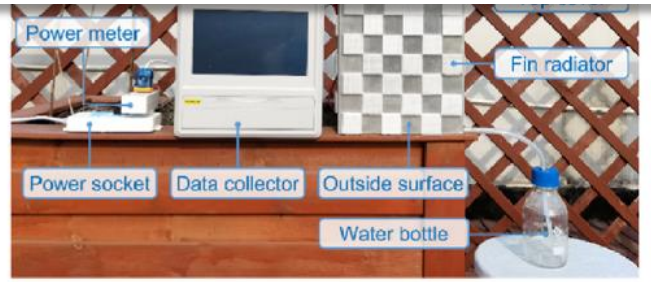
4.5 Waste heat recovery for extracting water from air

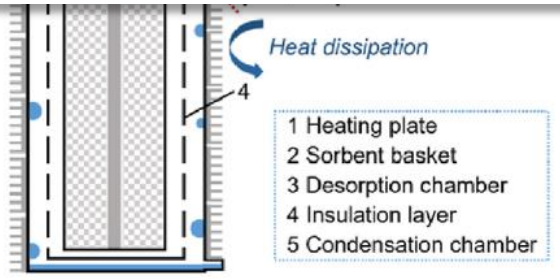
❑ Data center waste heat for atmospheric water harvesting by sorption-based method

- **Reference [1]**: use electric heater for heating the material, specific energy consumption is 1.7 kWh /L under low humidity conditions.
- **Proposed (**Heat Pump + Waste Heat**):** estimated energy consumption drops to 0.57 kWh/L (use heat pump (HP) to upgrade data center waste heat from 50-60°C to 100°C)
- **Assumption**: Heat pump COP is 3

- *Baseline scenarios: condensation-based method, using grid electricity*
- *Reuse 1 MJ waste heat for water harvesting can reduce CO_2e of **13.7 – 83.8 g***





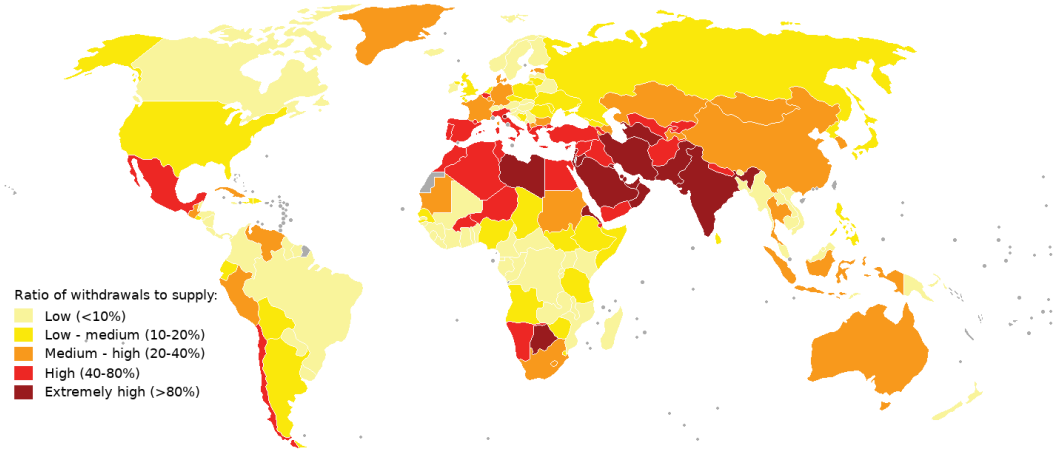


[1] Chen Z, Deng F, Yang X, et al. Highly efficient portable atmospheric water harvester with integrated structure design for high yield water production[J]. Energy, 2024, 293: 130749.
[2] Deng F, Poredoš P, Yu J, et al. Waste heat-driven water generator with optimized multi-cycle strategy for high water yield[J]. Applied Energy, 2024, 375: 124090.

4.5 Waste heat recovery for extracting water from air

❑ Beyond Energy Savings

- **Water Harvesting**: A 30 MW data center can support roughly the annual household water demand of approximately 800–1,000 homes.
- **Beyond Energy**: In arid regions, water resource is more valuable than simple electricity savings.
- **Opportunity**: Utilizing data center waste heat for atmospheric water harvesting represents a promising way to support water-stressed communities.



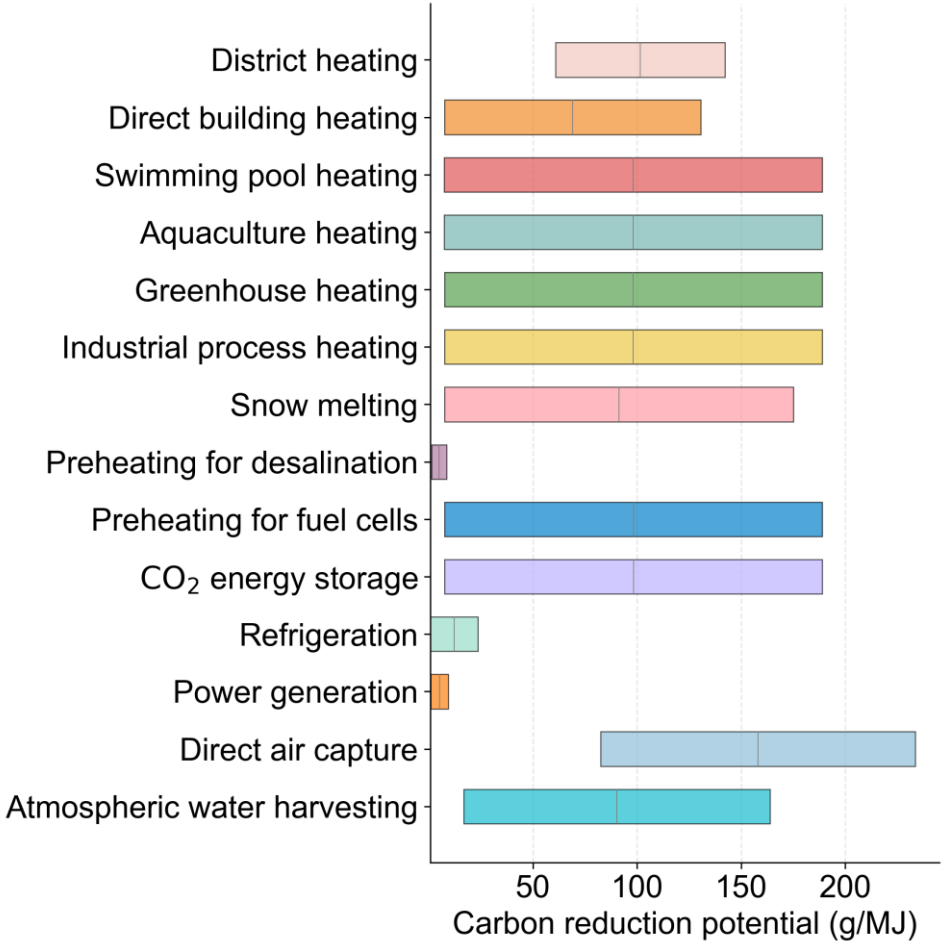
Map of global water stress in 2019 [1]



[1] Kummu M, Guillaume J H A, et al. The world's road to water scarcity: shortage and stress in the 20th century and pathways towards sustainability[J]. Scientific reports, 2016, 6(1): 38495.

5. Summary

❑ Carbon reduction potential



Discussion

- In real-world applications, supply–demand matching is critical and can significantly affect actual utilization efficiency.
- Traditional heating applications often face seasonal mismatches, as waste heat can typically only be utilized during the winter period.
- Technologies like Direct Air Capture and Atmospheric Water Harvesting can utilize waste heat continuously throughout the year.
- Actual impact is site-specific and requires a case-by-case analysis.

6. Call for action

- Problem to solve
  - Large amounts of AI data center waste heat are currently unused.
  - A systematic framework is needed to identify the best recovery pathways.
- Expected Outcomes
  - Optimal pathways for different data centers
  - Energy-saving and carbon-reduction potential
  - Economic and deployment assessment
- How to Get Involved
  - Share data and case studies, website interface
  - Collaborate on pilot projects
  - Contribute to an open white paper

Thank You!

